

Did Banks Pay Fair Returns to Taxpayers on TARP?

THOMAS FLANAGAN and AMIYATOSH PURNANANDAM*

ABSTRACT

Financial institutions received investments under the Troubled Asset Relief Program in a bad state of the world but repaid them in a relatively good state. We show that the recipients paid considerably lower returns to taxpayers compared to private-market securities with similar risk over the same investment horizon, resulting in a subsidy of over \$50 billion on the preferred equity investment by the government. Ex-post renegotiation of contract terms limited the upside gains received by taxpayers in good times and contributed to the subsidy. These findings have important implications for the design and implementation of future bailouts. Our simple methodology for calculating the subsidy can be applied to evaluate the financial costs of other bailouts.

BANKING CRISES ARE FAIRLY COMMON around the world. For instance, Laeven and Valencia (2008) identify 124 episodes of systemic banking crises across the world from 1970 to 2007. Laeven and Valencia (2008) further show that governments frequently respond to these crises by providing financial support, including the injection of fresh capital into the banking sector. More recently, multiple bank failures in 2023 were followed by the expansion of deposit insurance to cover private-sector losses. These stylized observations further underscore the importance of quantifying the economic costs of government interventions. In particular, how large a subsidy, if any, do banks receive through these interventions, and how does the implementation of these interventions affect the extent of the subsidy? Despite the importance of these questions for the design of future bailouts, there is little empirical research on the measurement of bailout subsidies on a risk-adjusted basis. We

*Thomas Flanagan is with Fisher College of Business, The Ohio State University. Amiyatosh Purnanandam is with Ross School of Business, University of Michigan. We thank Pat Akey, Juliane Begenau, Allen Berger, Stefan Nagel, Jonathan Parker, Paolo Pasquariello, Uday Rajan, Anil Shivdasani, Amir Sufi, Anjan Thakor, Neng Wang, and Zhenyu Wang, as well as participants at SFS Cavalcade 2021, WFA 2021, University of Toronto Finance Conference, Wisconsin Real Estate Research Conference, and seminars at the Bank of Portugal, University of Colorado at Boulder, Miami, PUC Chile, University of Virginia, and the University of Michigan for comments on the paper. We are grateful to Wei Xiong (Editor), the Associate Editor, and two anonymous referees for many insightful suggestions. We have read *The Journal of Finance* disclosure policy and have no conflicts of interest to disclose.

Correspondence: Thomas Flanagan, Fisher College of Business, The Ohio State University, Columbus, OH 43210, USA; e-mail: flanagan.224@osu.edu

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propose a simple model-free framework to do so using an attractive empirical setting: capital injections by the U.S. Treasury under the Troubled Asset Relief Program (TARP) in 2008 and 2009. We compute the subsidy as the difference between the realized return to taxpayers on TARP and a “fair” return that is measured as the return that a private investor would require to hold securities with similar risk over the same period.

A clear assessment of TARP’s financial cost is also important for informing policymakers and the public.¹ While the academic literature makes a clear distinction between an investment’s paper versus economic profit, bankers and some market observers often stress the fact that TARP recipients paid taxpayers back with profits, without making a clear distinction between the two.² The fact that these investments were paid back with positive returns has been used as an argument in favor of ex-post renegotiation of TARP contract terms, including early repayment and repurchase of warrants by the recipients, and broader policy changes in the banking sector after the recovery from the financial crisis. Such a narrative misses a critical aspect of finance theory, namely, that the fair return on an investment must compensate for the underlying risk of the security. Because TARP investments were made in a bad state of the world, a lack of risk adjustment can lead to substantial underestimation of the subsidy.

We focus on a specific part of the TARP intervention, the Capital Purchase Program (CPP). Under this program, the Treasury invested \$205 billion in financial institutions through preferred equity with warrants. We observe the exact dates and amounts of CPP investments as well as their repayments, including the cash flows received from the exercise of the warrants. We compute the internal rate of return (IRR) of these cash flows for each recipient and label it “TARP’s return.” We compare TARP’s return to carefully selected benchmarks on other securities with comparable risk to estimate the extent of the subsidy.

Preferred equity issued under the TARP had the same seniority as the existing preferred equity of the recipient banks. Thus, market-based returns on the existing preferred equity of the same bank over the same time horizon provide close to an ideal benchmark. Our approach obviates the need to rely on an explicit asset pricing model—we simply compare returns on two securities with similar risk issued by the same firm over the same time horizon. While not all TARP recipients had actively traded preferred equity outstanding at the time of TARP investment, 20 of the largest recipients covering about 80% of total

¹ Public opinion plays an important role in designing and implementing banking policies, especially those related to bank failure and bailouts. For example, Brown and Dinc (2005) show that political concerns play a major role in government interventions in distressed banks. Agarwal et al. (2014) document the importance of the regulator’s incentives on the implementation of identical policy rules.

² For example, see “U.S. ends TARP with \$15.3 billion profit” at <https://money.cnn.com/2014/12/19/news/companies/government-bailouts-end/>. Similarly, Wells Fargo’s CEO John Stumpf stated that TARP’s “success also generated financial returns for taxpayers, including \$1.4 billion in dividends paid to the U.S. Treasury by Wells Fargo.”

investments made under the plan did. TARP recipients paid 11% annualized returns to the taxpayers compared to the benchmark's annualized return of 39% over exactly the same time horizon. On a bank-by-bank basis, 19 of the 20 TARP recipients paid lower returns than the benchmark. Our next benchmark is an index of preferred equity instruments issued by S&P 500 firms.³ We are able to compute the subsidy for the entire sample of banks using this benchmark. On a value-weighted basis, TARP's return was 9% per annum compared to 39% for the benchmark. These estimates translate into a subsidy of \$54 billion on an aggregate basis.

One may be concerned that preferred equity invested under TARP came with warrants, whereas our benchmark securities are plain preferred equity. Since warrants are riskier than preferred equity, the return on plain preferred equity provides a conservative estimate for the cost of capital of TARP (e.g., Coval and Shumway (2001)). Further, the presence of warrants is unlikely to make a significant difference in the risk of these claims because ex-ante valuation estimates suggest that the warrants represented only about 8% to 10% of the value of the total investments (see Veronesi and Zingales (2010)). Notwithstanding, we directly address this issue by creating a synthetic benchmark that replicates the returns to a portfolio that invests 90% in the preferred equity of the bank and 10% in an American Call Option that closely mimics the value of warrants based on the TARP's terms. Our results remain similar.

We also consider as a benchmark the return earned by a large private investor, Warren Buffett, on his investment in Goldman Sachs during the financial crisis. Although the Buffett deal had the same security design, namely, preferred equity with warrants, it had a much higher coupon rate (10% against the CPP's rate of 5%), a prepayment penalty, and a significantly higher number of warrants attached to it. Compared to this benchmark, the TARP's return was even lower—50% lower on an annualized basis.

In the next part of the paper, we focus on the ex-post renegotiation of TARP contract terms. As per the original terms of TARP, preferred equity could not be paid back before a maturity of three years, and the warrants had a maturity of 10 years. However, banks actively renegotiated these terms soon after the economy began to recover. On average, preferred equity investments were repaid almost a year earlier than the three-year maturity initially agreed upon. The warrants were, subsequently, either repurchased by the issuing bank at a model-based valuation or sold in an auction.

The renegotiated terms had the potential to lower taxpayer returns through both unattractive repayment terms of the preferred equity and lower valuation of the warrants. Based on prevailing market prices, the yield on comparable preferred equity at the time of the TARP prepayment was about 4%, that is, lower than the 5% coupon rate on TARP securities. Since prepayments were

³ Although we miss bank-specific variation in returns in this benchmark, it is still an attractive benchmark for the TARP investment because about 70% of issuers in this index are financial institutions, and during the sample period, time-series variation in returns dominates differences in returns across banks.

made at par value, early repayment amounted to a subsidy of approximately \$2 billion or, equivalently, about 1% of the investment amount.

To calculate the effect of warrant disposal, we take a model-free approach since estimation of forward volatility on long-dated securities is highly sensitive to modeling assumptions. Specifically, we analyze how the recipient banks' stock prices reacted to the news of warrant disposition on two important dates: the announcement on June 26, 2009 by the Treasury detailing its framework to repurchase warrants, and the date on which a bank received approval for the disposal of warrants, which varied by the bank. Consistent with the view that warrant repurchase was beneficial to banks, their stock price experienced a +1.7% cumulative abnormal return (CAR) in a three-day window surrounding the Treasury's announcement of the framework. Since warrants were either sold in the market based on an auction or repurchased by the bailed-out institutions at a negotiated price, we are able to tease out the impact of renegotiation more carefully. The repurchased warrants represent successful renegotiation by the banks, whereas the sold warrants do not. Banks whose warrants were sold in the market had significantly lower stock returns in a three-day window surrounding the warrant disposal date compared to banks whose warrants were repurchased.

Although a detailed analysis of the causal effects of renegotiation on gains to shareholders and managers is beyond the scope of our paper, we do provide some suggestive evidence on this front in the last part of the paper. We show that banks with larger subsidies increased their dividend payments soon after the repayment, and bank CEOs gained significantly in terms of their total compensation. Further, CEOs who were able to successfully repurchase warrants from the Treasury earned significantly higher compensation than CEOs of banks whose warrants were sold in an auction. In conclusion, we show that taxpayers received considerably lower returns, whereas shareholders and managers benefitted from TARP. Further, gains from recovery were not shared with taxpayers in the same amount as originally promised.

Our results have immediate implications for the disclosure and computation of bailout subsidies, the design of future bailouts, and theoretical models of security design. The Congressional Budget Office (CBO) is tasked with calculating the financial performance of TARP investments. They use realized gains and losses on these instruments and the risk-free rate to compute their net present value (NPV). As per their estimate, CPP resulted in a net gain of \$16 billion, whereas our estimates show a subsidy of over \$50 billion.⁴

Our results show that the implementation of bailouts must account for the opportunity cost of penalty-free prepayment and warrant renegotiations. Taxpayers are likely to earn a higher return if the prepayment of investment in the good state of the world comes with a penalty. Further, their returns are likely to be higher if warrants are sold in an auction rather than repurchased at a model-determined price. While the economic rationale for financial assistance at the time of crisis is clear, there is no clear economic reason for

⁴ The report is available at <https://www.cbo.gov/publication/57341>.

providing banks with a subsidy once financial markets have recovered. Within the framework of a Bagehot-style bailout, it is not surprising that TARP's pricing was favorable to the banks compared to prevailing private-market benchmarks such as the Buffett deal. Yet, a Bagehot approach does not advocate for ex-post renegotiations to allow for early repayment or repurchase of warrants at favorable terms. Minimizing ex-post subsidies to banks should be an important component of any future bailout package.

A suitably designed security for bailouts, as in the theoretical model of Philippon and Schnabl (2013), requires participating banks to share their upside gains with the government in good times, for example, through adequately priced warrants. Our findings show that even though the securities were designed as such, ex-post renegotiation limited taxpayers' upside gains. Further, Bianchi (2016) and Dávila and Walther (2020) show that the possibility of a bailout changes the ex-ante leverage or risk-taking incentives of banks. Since highly levered and riskier banks are more likely to do better in good times, their warrants are more likely to have higher value when the economy recovers. Taxpayers may therefore be at a severe disadvantage if these warrants are renegotiated in good times: highly levered institutions receive higher subsidies at the time of the bailout, and the leverage gain is not shared with taxpayers in the upside. Therefore, future theoretical models of bailout design must take seriously into account the government's commitment not to renegotiate these contracts, and the renegotiation-proofness of bailout design in general (Hart and Moore (1988), Bolton (1990)). Even if such a contract cannot be written, our results show that reliance on a market-based mechanism for the ex-post disposition of securities can significantly limit the cost to the taxpayers.

A large academic literature on TARP is covered in survey papers such as Berger and Roman (2020) and Calomiris and Khan (2015). However, no previous work has analyzed the returns on TARP securities compared to market benchmarks, and the effect of renegotiations on these returns. We discuss the connection between our work and prior literature in detail in Section I. The remainder of the paper is organized as follows. In Section II, we develop our framework. Section III summarizes our data. Section IV presents the results. Finally, Section V concludes.

I. TARP Details and Literature Review

TARP was one of the key interventions undertaken by the U.S. Treasury to stabilize U.S. financial firms after the collapse of Lehman Brothers in September 2008.⁵ Under the overarching umbrella of TARP were several specific programs, such as CPP, Asset Guarantee Program (AGP), Auto-sector bailout, and investment in AIG. Our focus in this paper is on the CPP, which injected capital into the banking system in the immediate aftermath of the crisis.

Under the CPP, the U.S. Treasury invested \$205 billion in about 700 financial institutions in the country. Most such investments occurred in the last

⁵ See <https://www.treasury.gov/initiatives/financial-stability/about-tarp/Pages/default.aspx>.

quarter of 2008, but they continued into the early part of 2009 as well. These investments were structured as preferred equity with warrants that gave the Treasury an option to buy common equity of the firms anytime in the future up to 10 years after the investment date. The coupon rate was set at 5% per annum for the first five years of investment, and 9% thereafter. Warrants allowed the Treasury to purchase common shares equal in cost to 15% of the amount invested at a strike price set at the average of the institution's common stock price over the 20 trading days preceding the date of investment.⁶

The preferred equity was perpetual in nature, with the condition that it could not be redeemed before a period of three years. However, the American Recovery and Reinvestment Act of 2009 (ARRA) amended TARP to allow for early repayment of the preferred equity investment subject to Federal approval. Following the change, in June 2009, the Federal Reserve laid out criteria under which banks could repay TARP before the original maturity date.⁷ Of the \$205 billion in CPP investments, 90% of the principal was repaid before the original maturity. Soon after the repayment of preferred equity, outstanding warrants were disposed of by the Treasury for a vast majority of banks.

A vast literature on TARP is covered comprehensively in Berger and Roman (2020) and Calomiris and Khan (2015). Berger and Roman (2015), Li (2013), Bayazitova and Shivdasani (2012), Black and Hazelwood (2013), Duchin and Sosyura (2014), Chavaz and Rose (2019), and Berger, Roman, and Sedunov (2020), among others, study the effect of TARP on outcomes such as credit supply, bank competition, and risk-taking. Veronesi and Zingales (2010) and Glasserman and Wang (2011) focus on the valuation of TARP instruments and bank shareholders.

Our study complements Veronesi and Zingales (2010), who analyze changes in the value of the financial claims of the 10 largest recipients of TARP around the announcement date of the CPP program. Our work is conceptually related to Lucas (2019), who advocates the use of state contingent pricing in computing the fair value of bailout assistance. Methodologically, our work is related to Begenau, Piazzesi, and Schneider (2015) and Begenau and Stafford (2019), who use a replicating portfolio approach to uncover banks' risk exposure and valuation. Our findings are also related to Wilson (2009), who shows that the TARP warrants were undervalued when Old National Bancorp, the first publicly traded company to do so, repurchased it from the Treasury. Unlike these studies, we focus on ex-post realized returns of preferred equity

⁶ Naturally, the arrangement to buy common shares based on the average share price of the bank was applicable only to publicly traded institutions. These institutions received over 97% of the amount invested under the CPP.

⁷ As stated in the Department of Treasury's Warrant Disposal Report dated December 31, 2012, "Originally, the CPP contracts provided that an institution could not redeem the investment within the first three years except with the proceeds of a "qualified equity offering" (QEO), which is an offering of securities that would qualify as Tier 1 capital. The repayment terms of the contracts were later effectively amended by the ARRA, which provides that an institution can repay from any source of funds and without regard to any waiting period."

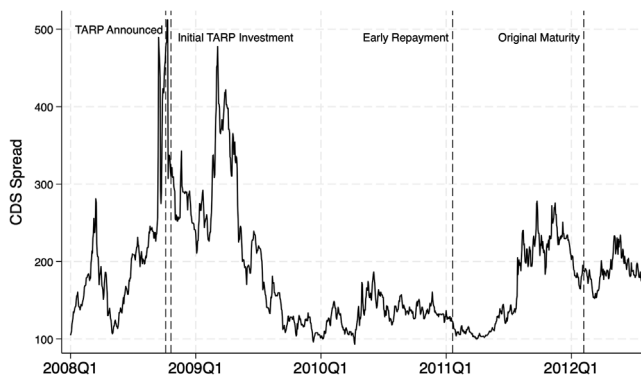


Figure 1. Bank CDS spread. This figure plots banks' CDS spreads over the TARP CPP investment horizon. Bank Index CDS Spread is measured by the DS North America Banks 5-Year Credit Default Swap Index. The dotted line labeled TARP Announcement corresponds to the TARP announcement on October 13, 2008. Initial TARP Investment is the date of the first TARP investment date on October 28, 2008. The dotted line labeled Original Maturity is the average earliest repayment date stipulated in the original TARP CPP Term Sheet. The dotted line labeled Early Repayment is the average TARP CPP repayment date conditional on repaying before the original maturity.

based on actual repayments.⁸ This approach allows us to shed light on the risk-return trade-off for TARP investments and directly assess whether the recipients enjoyed a subsidy or not in a model-free approach using ex-post cash flows after accounting for the renegotiated terms. These findings have not been previously documented to the best of our knowledge. Finally, our study also relates to the literature on the role of government support to the banking sector such as Atkeson et al. (2019) and Gandhi and Lustig (2015). The subsidy enjoyed by the shareholders of the bank in bad times can be an important driver of bank valuation and returns in good times.

II. Our Framework

A key feature of the CPP investments is that they were made at a time of great distress in the overall economy and specifically in the financial sector. U.S. banks' equity return dropped by almost 25% between January 1, 2008 to September 20, 2008, right after the collapse of Lehman Brothers. Banks' default risk increased by a significant amount as evidenced by a 312% increase in their credit default swap (CDS) spreads (see Figure 1). The short-term debt market froze due to disruptions in money market mutual funds (see Kacperczyk and Schnabl (2010)). Overall, this was an extremely bad state of the world, and private investors were reluctant to provide capital to the banking

⁸ For instance, the preferred equity valuation in Veronesi and Zingales (2010) assumes that preferred equity was fully repaid after five years. Ex post, this was renegotiated and the largest banks repaid the preferred equity in less than two years.

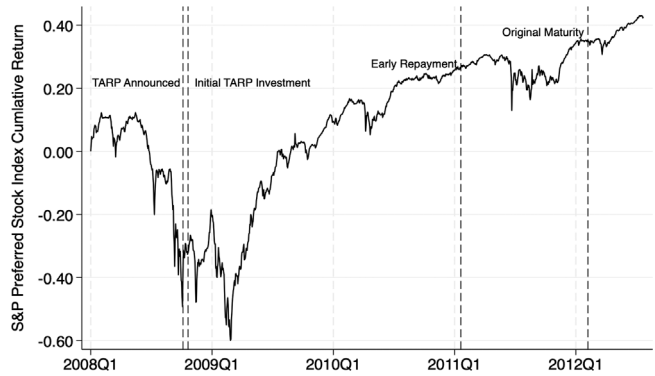


Figure 2. Benchmark preferred equity index. This figure plots the daily cumulative returns realized by the S&P U.S. Preferred Stock Index over the TARP CPP investment horizon. The cumulative return is normalized to zero on January 1, 2008.

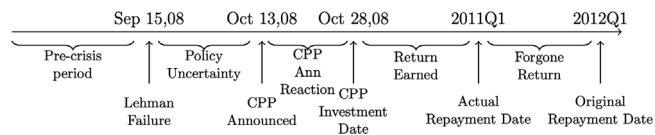


Figure 3. TARP repayment timeline.

sector. At a time like this, the required rate of return from a market participant's perspective is likely to be high to compensate them for the additional risk they are taking (see Figure 2).

We measure the “fair” return on TARP investments as the rate of return that a replicating portfolio in financial markets would earn. Our study therefore analyzes whether recipients paid the Treasury the same rate that they would have paid to raise similar funds from the private markets on the same day. The following timeline (Figure 3) presents the dates of key events during this period.

In the immediate aftermath of the failure of Lehman Brothers in September 2008, there was considerable uncertainty about the policy intervention as well as the precise nature of the policy (e.g., asset purchase versus equity injection in banks; see Calomiris and Khan (2015)). Eventually, on October 13, 2008, the CPP announcement was made public, and it became clear that the Treasury would invest in financial firms using preferred equity. Actual investment began two weeks later when the top nine banks of the country received CPP investments on October 28, 2008. A number of other institutions received funds in the subsequent weeks and months. These funds were repaid, on average, in the first quarter of 2011, compared to the original earliest repayment date of the first quarter of 2012. We measure returns from the date of investment (e.g., October 28, 2008 in the timeline above) to the date of actual repayment (indicated by 2011Q1 in the timeline above). The actual investment date and

repayment date vary for each bank depending on the exact date they received the funds and paid the funds back.

Using the Treasury's data set on TARP, we obtain daily cash flows from every TARP investment, including investments, dividends, principal repayments, and warrant proceeds. The TARP return for each bank is computed simply as the annualized IRR that sets the NPV of these cash flows to zero. Benchmark returns are calculated as the annualized rate of return from investing in the respective market benchmark using the same investment horizon, that is, from the date of investment to the repayment of each bank's TARP investment. The [Appendix](#) provides an example of this computation using investment in JP Morgan Chase as an example.⁹

Between September 15 and October 13, labeled the “policy uncertainty” period in the timeline, private markets faced uncertainty about the policy intervention itself. Part of the subsidy that banks enjoyed came from the effect of policy intervention on security prices after the announcement of the CPP on October 13. The other part comes from receiving cheaper funds from the Treasury conditional on the survival of the entire financial system. As the timeline shows, our benchmark returns begin two weeks after the policy announcement. During the “CPP Ann Period” (October 13 to October 28), market prices had sufficient time to adjust to the news of policy intervention. If the policy announcement itself resolves the uncertainty about the aggregate stability of the system (e.g., Roch and Uhlig (2018)), then our measure captures the subsidy due solely to the differential pricing of the standard risk conditional on the resolution of uncertainty about the financial stability of the entire sector. As Veronesi and Zingales (2010) show, market prices did indeed show a considerable response to the announcement of CPP right after October 13. Consistent with their findings, bank CDS spreads dropped from 512 basis points (bps) to 309 bps during the “CPP Ann Period” (see Figure 1).

It can be argued that some uncertainty about policy interventions remained even after October 28. To the extent that was the case, our subsidy measure includes some effects of policy uncertainty as well. We do not separate these two parts of the subsidy. However, in an additional test, we compare TARP returns with returns on private-market securities in periods when uncertainty about the aggregate stability of the financial sector had decreased or even been resolved. Specifically, we compare the TARP investment return with the return on private-market securities shifted by a period of six months to one year.

A. Limitations of Our Framework

Our framework has certain limitations. First, we do not capture welfare gains from TARP. Therefore, our subsidy measure is specific to financial gains and losses alone. Second, there may be a concern about the lack of liquidity in the private-market benchmarks that we use for the calculation of fair returns.

⁹ Some banks received TARP investments on two different dates. For such banks, we compute a value-weighted portfolio return proportional to the investment amounts.

And third, while our benchmark return for the preferred equity part is direct, for the warrant part of the benchmark return, we need to rely on a model. We address issues regarding the lack of liquidity and the need for a warrant valuation model later in the paper, after we present our main results. In general, we show that the economic magnitude of these limitations is not large enough to overturn the key message of our paper. Finally, our study computes the subsidy without considering the endogeneity concern whereby the benchmark return in the private market may itself depend on the extent and nature of government intervention. In this sense, our estimate is “marginal” in nature, that is, it computes the subsidy on the marginal taxpayer dollar conditional on the government’s desire to intervene.

III. Data

We collect data from several sources. Information on TARP investments and repayment comes from the Department of Treasury’s website.¹⁰ This data set provides bank-level information on the size, timing, and repayment of CPP investments including dividends and warrant proceeds. We limit the sample to banks receiving preferred equity investments, which make up over 99% of the total CPP investment amount.¹¹ We obtain information on warrant disposition, that is, sales and auctions, from the Treasury’s website as well.¹²

We obtain information on preferred equity benchmarks from two sources: Mergent FISD and Refinitiv Datastream. Mergent FISD provides information on the characteristics of preferred equity issued by the recipient banks. From this data set, we take the most recent preferred equity issued prior to 2008 with pricing data available until the end of the original TARP maturity for our benchmark returns. Refinitiv Datastream provides information on returns. For the bond benchmark, we obtain data from Bond CRSP Link provided by WRDS.¹³ Bond returns are calculated using information on bond coupons from the FISD and market prices from the TRACE data sets.¹⁴

We obtain bank-level common equity returns from CRSP. We use the Bank CRSP-Compustat merged data set to obtain data on dividends and share

¹⁰ Available for download at <https://www.treasury.gov/initiatives/financial-stability/reports/Pages/TARP-Investment-Program-Transaction-Reports.aspx>.

¹¹ Some smaller banks received subordinated debt investments. We exclude these investments from our analysis to maintain uniformity in benchmarks and to facilitate comparison of TARP subsidies across banks.

¹² Available at <https://www.treasury.gov/initiatives/financial-stability/reports/Pages/Warrant-Disposition-Reports.aspx>.

¹³ We consider bonds issued before the CPP investment date and maturing after the redemption date. We also impose standard filters used in the corporate bond literature. Specifically, we restrict the sample to senior bonds, corporate debentures and medium-term notes, bonds with \$1000 principal amount, bonds with nonchanging coupon, and bonds with greater than \$50,000,000 offering amount.

¹⁴ Bond returns are computed at a monthly frequency due to limited daily trading. We convert monthly returns to daily returns when calculating the returns over a given TARP investment horizon. Returns on preferred equity benchmarks are available on a daily basis.

repurchases.¹⁵ Executive compensation data come from Execucomp. Using the Fed NY link file, we link banks in CRSP to Bank Holding Company from the Call Reports for bank characteristics. We measure bank characteristics in 2009Q1, the average initial investment period for TARP CPP investments. Finally, we obtain information on the implied volatility of the longest dated option of the recipient banks using the OptionMetric database.

Table IA.I in the [Internet Appendix](#) provides further details on the sample used in our analysis.¹⁶ We start with all 707 banks that received an investment from the TARP's CPP program. Of these banks, 653 received preferred equity investments. We have individual preferred equity returns for 20 banks. Finally, we obtain individual senior bank bond returns for 21 banks, 17 of which overlap with the preferred equity subsample. While the samples requiring returns on individual bank preferred equity or bonds are much smaller, they both account for almost 80% of the CPP's investment in dollar value terms.

A. Descriptive Statistics

Table I provides key summary statistics for TARP investments and bank characteristics. As shown in the table, the average bank received \$312.9 million under the CPP. The earliest TARP investment was made in October 2008, and the median investment date was January 2009. While the average effective maturity of the investments was just over three years, about half of the 653 banks repaid the principal investment before the original maturity. Larger recipients paid it back much earlier. On a value-weighted basis, the effective maturity was 1.7 years.

IV. Results

In this section, we present our estimates of TARP's risk-adjusted realized return for various benchmarks.

A. Estimation of Benchmark Returns

A.1. Same Bank Preferred Equity Return

Preferred equity issued under the CPP had the same seniority in the bank's capital structure as existing preferred equity of the bank.¹⁷ Thus, the closest market-based benchmark comes from the traded preferred equity of the same bank outstanding at the time of CPP's investment. This benchmark captures

¹⁵ A few banks in CRSP received multiple TARP investments. For these banks, we keep the largest Treasury investment for analysis requiring CRSP data.

¹⁶ The [Internet Appendix](#) is available in the online version of the article on *The Journal of Finance* website.

¹⁷ As quoted in the congressional oversight panel report dated July 10, 2009 "The CPP preferred shares.... are senior to the institution's common shares, have an equal preference to existing preferred shares, and are non-voting."

Table I
Summary Statistics

This table presents summary statistics for banks receiving TARP CPP investments. TARP Investment Amt. is the total investment a bank received through the TARP CPP. Investment Date is the year-month in which a bank's first TARP investment was made. Investment Maturity is the number of years from the initial investment until the bank both repaid the TARP investment principal and repurchased/sold its warrant. EW Investment Maturity is the equal-weighted maturity across all banks. VW Investment Maturity is value-weighted maturity using the CPP investment amount as the weight. TARP Cash Back is the total cash received by the Treasury under the program.

<i>N</i>	Mean	<i>SD</i>	Min	P25	P50	P75	Max	<i>N</i>
TARP Investment Amt. (\$ Millions)	312.90	2,100.90	0.30	5.00	11.73	35.00	25,000.00	653
Investment Date	2009m1	2.64	2008m10	2008m12	2009m1	2009m3	2009m12	653
EW Investment Maturity (Years Until Warrant Disposal)	3.39	1.64	0.24	2.37	3.29	4.34	9.49	653
VW Investment Maturity (Years Until Warrant Disposal)	1.71	1.17	0.24	1.13	1.35	2.24	9.49	653
TARP Cash Back (\$ Millions)	346.24	2,387.24	0.00	4.68	11.94	33.82	32,839.27	653

Table II
Benchmark Returns

This table compares TARP returns to various public market benchmarks. TARP Return Realized is calculated as the annualized IRR of TARP cash flows. TARP Subsidy is the difference between a benchmark index return and TARP's return. EW Return is the equal-weighted return across banks. VW Return is the value-weighted return using the original TARP investment amount as the weight. S&P U.S. Preferred Stock Index Return Realized is the annualized rate of return from investing in the S&P U.S. Preferred Stock Index for the same investment horizon as the bank's TARP investment. Same Bank Preferred Equity Return Realized is computed using the same bank's preferred equity returns and the same investment horizon as that bank's TARP investment. Preferred Equity With Warrant Return Realized is computed using the returns to a replicating portfolio with a 90% weight to preferred equity claims and 10% to warrants. The warrant returns are calculated following the Treasury's methodology of valuing the claims as an American Call Option on a dividend paying stock using a binomial approximation of the Black-Scholes model. The TARP with Buffett Terms benchmark is computed by taking the IRR of a bank's cash flows if the TARP investment had the same terms as Buffett's investment in Goldman Sachs. Same-Bank Senior Bond Return Realized is computed using the same bank's senior bond returns. Individual bank preferred equity returns are taken from Refinitiv Datastream using the most recent preferred equity issue prior to 2008 with trading data available until the end of the original TARP maturity. *t*-Statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	EW Return		VW Return		Obs.
Panel A: Same-Bank Preferred Equity					
TARP Return Realized (Same-Bank Sample)	0.10***	(9.66)	0.11***	(9.90)	20
Individual Bank Preferred Equity Return Realized	0.49***	(7.84)	0.40***	(7.42)	20
TARP Subsidy (Difference)	0.39***	(6.61)	0.29***	(6.20)	20
Panel B: Preferred Equity Index					
TARP Return Realized	−0.01	(−0.82)	0.09***	(15.21)	653
S&P Preferred Stock Index Return Realized	0.27***	(37.80)	0.39***	(70.85)	653
TARP Subsidy (Difference)	0.28***	(25.16)	0.30***	(39.68)	653
Panel C: Pref. Equity with Warrant					
TARP Return Realized (Same-Bank Replicating)	0.10***	(9.33)	0.11***	(9.63)	19
Preferred Equity With Warrant Return Realized	0.50***	(7.34)	0.40***	(6.26)	19
TARP Subsidy (Difference)	0.39***	(6.33)	0.29***	(5.19)	19
Panel D: TARP with Buffett Terms					
TARP Return Realized	−0.01	(−0.82)	0.09***	(15.21)	653
Buffett IRR	0.23***	(13.25)	0.59***	(27.30)	653
TARP Subsidy (Difference)	0.24***	(18.54)	0.50***	(27.47)	653
Panel E: Same-Bank Bond					
TARP Return Realized (Same-Bank Bond Sample)	0.12***	(8.50)	0.12***	(9.34)	21
Individual Bank Senior Bond Return Realized	0.19***	(6.78)	0.20***	(6.61)	21
TARP Subsidy (Difference)	0.07***	(2.90)	0.08***	(3.29)	21

both the timing of the TARP investment horizon as well as the (time-varying) risk characteristics specific to the bank receiving the investment.

We are able to obtain these data for 20 banks in our sample, representing 78% of total CPP investment. Data on the total return of these instruments are available at the daily frequency, which allows us to compute the benchmark return for exactly the same time horizon as the CPP investment. Panel A of Table II provides the estimation results. Individual preferred equity holders

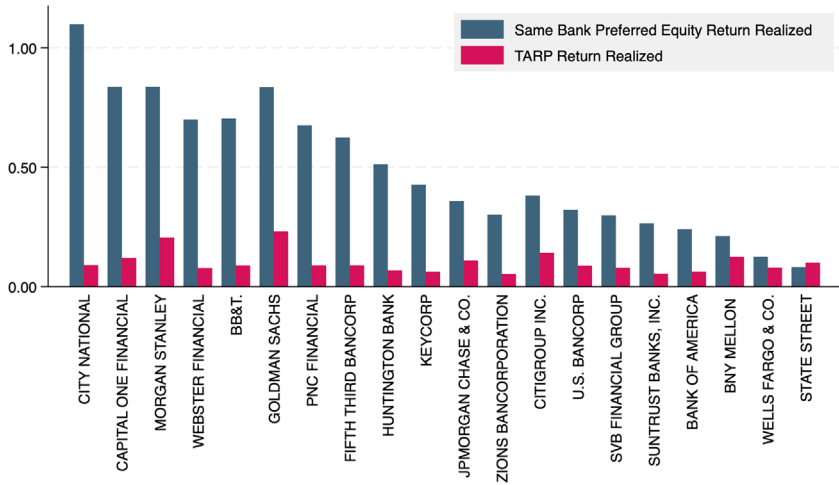


Figure 4. TARP investment versus same bank preferred equity. This figure compares a bank's TARP CPP return to the same bank's preferred equity return using the same buy and sell horizon as the TARP investment. Both returns are annualized. (Color figure can be viewed at wileyonlinelibrary.com)

earned a value-weighted annualized return of 40%, compared to the 11% earned by TARP.¹⁸ The equal-weighted results are similar, with a preferred equity return of 49% compared to 10% earned by TARP. Both differences are statistically significant at the 1% level. Figure 4 depicts the bank-level return on TARP investment and the bank's individual preferred equity. The difference between the TARP return and the preferred equity return is stark: across 20 banks, TARP earned considerably lower returns than preferred equity holders in 19 cases.

A.2. S&P U.S. Preferred Equity Index Return

A limitation of the same bank preferred equity benchmark is that it is not available for most firms. We use the return on S&P U.S. Preferred Stock Index as an alternative benchmark to estimate the subsidy for the entire cross-section of banks. This index is designed as an “investable benchmark representing the U.S. preferred stock market.” The bulk of the index constituents, 70%, correspond to the financial sector. For instance, the top 10 constituents include preferred shares from Citigroup, Ally Bank, JP Morgan Chase, Wells Fargo, and PNC Financial. The return on this index closely mimics the value-weighted return on preferred stocks for the 20 banks studied in the bank-level analysis. The two return series have a positive correlation of 95%. The index return therefore provides a useful benchmark for banks without traded preferred shares.

¹⁸ All value-weighted returns are computed with the original investment amount as the weight.

Panel B of Table II provides the estimation results. TARP's equal-weighted annualized return for the entire sample is -1% ; that is, on average, a bank in the sample returned less money than it received from TARP. However, the value-weighted return to the TARP investment is 9% per annum, which is very similar to the estimate based on the sample of 20 banks discussed above. Over the same period, the S&P U.S. Preferred Stock Index earned a much higher average annualized return of 39% . Thus, TARP's return is almost one-fourth of a comparable market investment. The subsidy of 30% is statistically significant at the 1% level.¹⁹

A.3. Replicating Portfolio Return

A shortcoming of the preferred equity return benchmarks is that we miss the warrant portion of the investment from the benchmark return. These warrants are American Call Options on the recipient bank's common stock. As such, they are riskier claims than both the common stock and preferred equity of the bank. Hence, the omission of warrants makes these benchmarks less risky, which biases against our finding.

To address this issue, our third benchmark replicates returns on a portfolio of preferred equity and warrants that closely mimics the value of TARP warrants. We closely follow the Treasury's methodology in valuing these warrants as an American Call Option on a dividend-paying stock using a binomial approximation of the Black-Scholes valuation model.²⁰ Using these model-implied values, we compute the return to the warrant part of the portfolio over the investment horizon. The replicating portfolio assigns a weight of 90% on the preferred equity claims and 10% on warrants, in line with the ex-ante valuation estimates of Veronesi and Zingales (2010). The Appendix provides further details on the warrant valuation methodology and assumptions underlying the model inputs.

Panel C of the table presents the results.²¹ The results are almost identical to those presented with straight preferred equity as the benchmark. This is not surprising because the portfolio has 90% weight on straight preferred equity and only 10% on warrants.

¹⁹ One potential concern with comparing TARP returns and traded preferred equity returns is that these instruments might have different exposures to duration risk. This feature is unlikely to be a significant issue for our results since ex-ante TARP investments had perpetual maturity, whereas traded preferred equity has an average duration of over 40 years. As discussed earlier, TARP recipients had the option to repay the investment after three years. In reality, however, they paid it even sooner. In Appendix A.2, we examine how the subsidy changes if we account for differences in duration assuming TARP investment has a shorter duration based on actual repayment. Our results become slightly stronger.

²⁰ Although TARP recipient banks had dividend restrictions during the repayment period, these restrictions limited *increases* in dividends, not the *level* of dividends.

²¹ We present this analysis for 19 banks for which we have preferred equity returns as well as actively traded options in the market. We do so because we need option-implied volatility for the estimation of the warrant valuation model.

A.4. Buffett's Investment in Goldman Sachs

Warren Buffett invested \$5 billion in Goldman Sachs soon after the failure of Lehman Brothers using the same security design as the TARP investment: preferred equity with attached warrants. However, the terms of his investment were considerably different. The Buffett deal had a 10% coupon compared to 5% for TARP, it had a prepayment penalty of 10% of the invested amount compared to none for TARP, and it came with a significantly higher number of warrants. Specifically, the Buffett deal gave his firm an option to buy common shares worth \$5 billion on the date of investment (i.e., 100% of the investment amount). The corresponding amount of common shares for TARP was set at 15% of the invested amount. This deal from a private-market investor provides us with an attractive benchmark since it closely matches the timing of the investment as well as the design of the security. Further, this benchmark is free from any liquidity-related issues that may contaminate our findings based on returns from the market instruments. We compute the benchmark return for each bank under the condition that they all received the same terms as the Buffett deal. Panel D presents the results. For this benchmarking, the subsidy is even higher at 50% per annum on a value-weighted basis.

The overall message from our benchmarking analysis is clear: taxpayers received at least 30% lower returns on an annualized basis compared to securities of similar risk.

A.5. Same Bank Senior Bond Return

Our next benchmark is constructed to capture a lower bound on preferred equity returns—the senior bond returns of the same bank. We have data on bond returns for 21 banks in the sample; these 21 banks account for 80% of the CPP's \$205 billion investment. Panel E of Table II shows that the bank's bondholders earned a value-weighted annualized return of 20% compared to the 12% earned by TARP. Thus, taxpayers earned almost one-half the return earned by senior claimholders in the same firm.

Our benchmarking analysis is based on realized returns. Junior claims are expected to earn higher returns than senior claims on average, but not in every sampling period. However, this does not pose any problem for our analysis since we analyze the return differential across junior and senior claims in a setting in which the investment was made in a very bad state of the world while the return was realized in a good state of the world. For such an investment horizon, the realized return on a junior claim should be especially higher. But that was not the case for TARP investments.

Table IA.IV in the [Internet Appendix](#) uses two bond indices as the benchmark return: (i) Bloomberg Barclays Investment Grade Bond Financial Institution and (ii) Bloomberg Barclays BBB Bond Index. The advantage of these benchmarks is that we are able to compare the returns for the entire sample of financial institutions, and not only the banks that have issued bonds. The results are similar.

Table III
NPV of TARP Investments

This table presents statistics on the NPV of TARP investments. Panel A presents the NPV for the full bank sample. For the full bank sample, NPV is calculated by discounting back the net cash flows from TARP investments using the cumulative returns on the S&P U.S. Preferred Stock Index. NP/Investment and NP/Assets normalizes Dollar NPV of TARP using the TARP investment amount and total bank assets, respectively. Panel B calculates NPV for individual banks with traded preferred equity. For the preferred equity bank sample, NPV is calculated by discounting back the cash flows from TARP investments using the cumulative returns on the same bank's preferred equity. Note that a negative NPV corresponds to a positive TARP subsidy.

Panel A: Full Sample								
	Mean	<i>SD</i>	Min	P25	P50	P75	Max	<i>N</i>
NPV of TARP (\$ Billions)	−0.083	0.531	−7.707	−0.014	−0.005	−0.002	0.093	653
NPV/Investment	−0.435	0.228	−1.030	−0.535	−0.400	−0.320	0.297	653
NPV/Assets	−0.009	0.009	−0.107	−0.011	−0.008	−0.005	0.005	197
Panel B: Same Bank Preferred Equity Sample								
NPV of TARP (\$ Billions)	−1.701	2.007	−8.860	−2.395	−1.184	−0.293	−0.003	20
NPV/Investment	−0.306	0.194	−0.596	−0.495	−0.309	−0.117	−0.001	20
NPV/Assets	−0.007	0.005	−0.016	−0.013	−0.005	−0.002	−0.000	20

B. Dollar Value of the Subsidy and Cross-Sectional Determinants

We estimate the dollar gains or losses on TARP investments by computing their NPVs. We do so by discounting the TARP cash flows at a rate that equals the benchmark returns calculated earlier. Results are presented in Table III. Panel A calculates NPV for the entire sample of banks using the cumulative return on the S&P U.S. Preferred Stock Index return over the investment period as the discount rate. Panel B repeats the exercise for the smaller subsample of 20 banks, with the bank's own preferred equity return as the discount rate. As shown in Panel A, the average "Dollar NPV" of TARP is negative \$83 million, meaning that TARP investments lost an average of \$83 million across 653 banks. This puts the total dollar cost of subsidy at roughly \$54 billion on an investment amount of \$205 billion. Panel B of Table III presents results for the sample of banks that have traded preferred equity. The results are similar.

The table also reports NPV per dollar of TARP investment. On average a \$1 investment in TARP lost 43.5 cents. From the bank's perspective, NPV per dollar of bank asset works out to -0.009, which means that the economic subsidy amounted to 0.9% of the bank's assets on average, which is a significant amount when compared with the average return on assets in the banking industry.²²

We also investigate whether the cross-sectional variation in subsidy is related to individual banks' risk as measured by beta, historical volatility,

²² For example, the average quarterly return on assets for FDIC-insured institutions was 0.88% in 2011Q4. See <https://www.fdic.gov/bank/statistical/stats/2020dec/fdic.pdf>.

and size. Unsurprisingly, riskier banks received economically larger subsidies from TARP since the terms of investment did not change with the recipient's risk. These results are reported in the [Internet Appendix Table IA.II](#).

C. Renegotiation of TARP Contracts

CPP investments were renegotiated between the U.S. Treasury and the recipients soon after the financial markets began to recover. The renegotiations involved shortening of maturity as well as repurchases of warrants by the issuers. Early repayment allowed banks to pay back the preferred equity principal at par value after yields had fallen significantly since the peak of the crisis. The original contract stipulated that banks could only repurchase warrants after at least three years of the repayment of the preferred equity principal. Renegotiation also allowed for early warrant repurchase. As a result, many banks began repurchasing warrants from the Treasury starting in July 2009 after the early repayment of the preferred equity principal.²³

Table IV contains descriptive statistics on the renegotiation of TARP contracts. We find that 304 out of 653 financial institutions in our sample repaid TARP earlier than the original three year term. On a value-weighted basis, 90% of TARP investment was renegotiated. Turning to warrants, about 97% of investments on value-weighted basis came with warrants attached to them. The Treasury's warrant disposition report provides information on 95% of warrant investments, including information on how the Treasury disposed of the warrants. On a dollar basis, about half of the warrants were repurchased by the banks, while the other half was sold to market participants. We compute the effect of these renegotiations on TARP returns and present the results below.

C.1. Early Repayment Subsidy

Early repayment renegotiation allowed banks to repay instruments with 5% coupon without any prepayment penalty when yields on comparable securities had come down. Between the time of TARP investment and repayments, the risk-free rate declined significantly, the banking sector's default risk improved, and the risk premium came down. As a result, the effective yield was likely to be much lower at the time of repayment compared to the time of investment.

To estimate the repayment subsidy given to banks, we first compute the prevailing market yield on preferred equity at the time of repayment. We compute the average preferred equity yield based on the sample of banks for which we

²³ The original terms in the EESA (Emergency Economic Stabilization Act of 2008) required the Treasury to dispose of warrants when "the market [was] optimal for such assets, in order to maximize the value for taxpayers." The ARRA passed in February 2009 amended the EESA to require the Treasury to "liquidate" warrants after repayment. Although the contract stipulates that warrants must be repurchased at fair market value, market benchmarks for such instruments (call options with 10-year maturities) are scarcely available.

Table IV
Renegotiation of TARP Contracts

This table presents descriptive statistics on the renegotiation of TARP contracts. Repaid Early is an indicator equal to one if a bank repaid the preferred equity principal before the minimum contractual maturity of three years. Investment Maturity is the number of years that a bank took to repay the principal of TARP conditional on early repayment. Includes Common Stock Warrants is an indicator equal to one if the TARP investment included warrants on publicly traded common stock. Warrant Disposition Data Available is an indicator variable equal to one if the banks' warrant disposition information is included in the Treasury's 2012 warrant disposition report. Warrant Repurchased by Bank indicates that the Treasury repurchased the warrant from the bank at a negotiated price. Warrant Sold at Auction indicates that the Treasury sold the warrant at auction to market participants. The Value-Weighted column computes averages using the original TARP investment amount as the weight.

	Equal-Weighted		Value-Weighted	
	Mean	N	Mean	N
Early Repayment				
Repaid Early	0.466	653	0.904	653
Investment Maturity (Years Condo. on Early Repayment)	1.939	304	1.190	304
Warrant Disposition				
Includes Common Stock Warrants	0.452	653	0.980	653
Warrant Disposition Data Available	0.245	653	0.953	653
Warrant Repurchased by Bank	0.756	160	0.493	160
Warrant Sold at Auction	0.244	160	0.507	160

have individual preferred equity data. Since the average duration for preferred equity is over 40 years while the TARP investment had a duration of only one to two years based on realized cash flows, we need to make some adjustments to account for the maturity difference. We do so in two steps. We first compute the spread on preferred equity as the difference between the preferred equity yield and the 30-year Treasury yield at the time of repayment.²⁴ In the second step, we add the spread to the two-year Treasury yield to estimate the preferred equity yield in the market at the time.

We compute the repayment subsidy as the NPV of the originally promised cash flows until three years from the investment date discounted at the prevailing preferred equity yield. We divide the estimate by the par value of the TARP amount to obtain a percentage measure of subsidy for each recipient.²⁵ Table V presents the results. We find that the average preferred equity yield in the market is 4.05% at the time of TARP repayment, compared to the 5%

²⁴ Although our preferred equity index has an average duration of 40 years, the closest publicly traded Treasury benchmark is a 30-year Treasury bond. We compute the average preferred equity yield on a daily basis using the sample of banks with fixed rate coupons with fixed maturities. Table IA.III in the [Internet Appendix](#) shows that the subsidies remain similar if we consider preferred equity with a floating rate coupon without subtracting the 30-year Treasury yield.

²⁵ A small fraction of banks (less than 2% on a value-weighted basis) repaid TARP in multiple installments. For these banks, we only compute the repayment subsidy using only the final payment installment. This does not qualitatively or quantitatively alter the results.

Table V
Early Repayment Subsidy

This table reports the early repayment subsidy banks received. Repayment Subsidy is the NPV of the remaining scheduled preferred equity cash flows (5% coupon + principal repayment at three years maturity) scaled by the investment amount of those cash flows. The discount rate, labeled as the (Adjusted) Perf. Equity Yield at Repayment, is taken as the market-based average preferred equity yield at the time of repayment, adjusted for differences in maturity with respect to the remaining TARP cash flows. Remaining Maturity at Repayment is the number of years at the time of repayment until the original three-year minimum maturity. The value-weighted column computes averages using the original TARP investment amount as the weight. *t*-Statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Equal-Weighted		Value-Weighted	
Repayment Subsidy	0.677***	(6.175)	1.051***	(16.417)
(Adjusted) Perf. Equity Yield at Repayment	4.050***	(70.733)	4.247***	(107.807)
Remaining Maturity at Repayment	1.061***	(23.457)	1.810***	(49.597)

coupon on TARP. On an equal-weighted basis, we find that the repayment subsidy is 0.7%. On a value-weighted basis, the subsidy is 1.1%. In other words, allowing banks to repay TARP early at par value had an opportunity cost of approximately \$2 billion based on the yield of comparable market securities at the time of repayment.

The subsidy that came from relatively attractive terms at the time of the bailout was provided to banks in a distressed state of the world. In contrast, the prepayment subsidy was provided after markets recovered. While providing a subsidy to avoid negative externalities in a crisis state is warranted, there is no clear economic justification for providing banks a subsidy after financial markets recover. These results provide an immediate policy recommendation for future bailout designs: prepayment must accompany a penalty if done in a good state of the world. Alternatively, these securities can be sold in an auction to better capture the present value of their expected future cash flows.

C.2. Warrant Disposition Renegotiation

We conduct an event study centered around the June 26, 2009 announcement of the Treasury's framework to repurchase warrants.²⁶ Under this framework, a bank can repurchase its warrant if the Treasury accepts its bid; otherwise, warrants are auctioned in the open market. If markets anticipate that the repurchases of TARP warrants are fairly priced, the announcement of this framework should have no effect on a bank's stock return. Alternatively, if common equity holders anticipate the terms of the warrant repurchases to be favorable, then there is likely to be a positive effect on stock returns. We test this hypothesis by measuring the bank's CARs using a (−1,+1) three-day window centered around June 26, 2009. To measure the economic value received by shareholders due to the warrant disposal, we value-weight the

²⁶ See www.treasury.gov/press-center/press-releases/Pages/200962612255225533.aspx

Table VI
Warrant Disposition Event Study

This table reports the bank stock price reactions to the announcement of bank repurchases of TARP warrants. In Panel A, the event study window is centered around a single day, June 26, 2009, when the Treasury announced its framework for repurchasing TARP warrants “quickly.” In Panel B, the event window is centered on the date when a bank announced its warrant transaction price and whether the warrant was repurchased by the bank or auctioned to market participants. Panel B reports CARs separately for banks with repurchased and auctioned (sold) warrants and tests the difference between them. Returns and CARs are calculated using a $(-1,1)$ three-day window using the CRSP VW index in a CAPM model. The mean returns are value-weighted by banks’ market equity, and standard errors are clustered at the bank level. t -Statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Warrant Repurchase Process Announced		
	(−1,1) Event Return	
Raw Bank Return	0.044***	(9.24)
CAR	0.017**	(2.23)
Observations	135	
Panel B: (Staggered) Warrant Transaction Price Disclosed		
	(−1,1) Event Return	
<i>Warrant Repurchased</i>		
Raw Bank Return	0.016**	(2.51)
CAR	0.002	(0.31)
Observations	101	
<i>Warrant Sold at Auction</i>		
Raw Bank Return	−0.013***	(−4.43)
CAR	−0.017***	(−3.51)
Observations	34	
<i>Difference</i>		
Raw Bank Return	0.029***	(4.09)
CAR	0.019**	(2.39)
Observations	135	

event-window stock returns using the market equity value, measured just before the event window, as the weight. We pool all banks with outstanding warrants together in this test because it was not clear which banks’ warrants would be repurchased or sold at the time. Next, we conduct a staggered event study around the date that a bank announced its warrant transaction price to the public and whether its warrant was repurchased or sold in an auction.

Table VI presents the results.²⁷ Panel A documents a positive and highly significant stock return of 4.4% around the June 26, 2009 event window. After

²⁷ In this analysis, the sample size for banks with warrant disposition information decreases from 160 to 135 because some banks do not have CRSP data coverage to compute abnormal returns. The betas used to compute abnormal returns are estimated from a monthly CAPM, and following Bayazitova and Shivdasani (2012), we estimate these parameters over January 2007 to September 2008 before any policy intervention.

adjusting for Capital Asset Pricing Model (CAPM) returns, the CAR remains a highly significant 1.7% during this window. Therefore, shareholders gained from the initial announcement of early warrant disposition.

Panel B presents the results of the staggered event study in which details of the warrant price and method of disposition were announced. Consistent with markets anticipating a repurchase subsidy from the June 26 announcement, we find no statistically significant abnormal return for banks whose warrants were repurchased. We do find, however, a statistically significant negative abnormal return of -1.7% for the subsample of banks whose warrants were sold at auction. Put differently, the positive returns that banks earned on the initial announcement date did not disappear if banks successfully bought their warrants back from the Treasury, but if they were unable to do so, the entire positive return was lost. This analysis reveals that bank shareholders captured some of the taxpayer gains of TARP investments via warrant repurchase at favorable terms.

To better understand the source of the warrant repurchase subsidy, we analyze the Treasury's methodology in determining "fair" repurchase prices.²⁸ The Treasury uses a binomial approximation to the Black-Scholes model. The key input in this valuation is the 10-year forward volatility, for which few, if any, market benchmarks exist. In practice, the Treasury took a linear interpolation between the implied volatility of a shorter-maturity option and the bank's 10-year historical volatility to arrive at its volatility estimate for estimating the warrant's fair value, but it used a winsorized version of the 10-year historical volatility. By cutting out tail observations, which might especially matter over a 10-year business cycle, the model could be significantly underestimating the true 10-year forward volatility.

In the next set of analyses, we compare the nonwinsorized 10-year historical volatility from CRSP to the implied volatility from the Treasury's warrant repurchase price.²⁹ We define *Volatility Subsidy* as the difference between the two volatility estimates. We also compute an alternate measure of *Volatility Subsidy* using the implied volatility from the longest dated traded call option from OptionMetric. If the term structure of volatility is upward-sloping, this should provide a market-based lower bound on the subsidy. Table VII presents the results. We find that the 10-year historical volatility from CRSP is 18.5% higher than the implied volatility used by the Treasury. The difference is statistically significant. Column (2) indicates that the option-implied volatility is 10.3% greater than the Treasury's implied volatility. In other words, banks likely repurchased these warrants at a favorable price from the Treasury since the volatility estimate is the key determinant of warrant valuation. We also

²⁸ The full report on the Treasury's warrant valuation methodology is available at <https://www.treasury.gov/initiatives/financial-stability/TARP-Programs/bank-investment-programs/cap/Documents/Jarrow%20TARP%20Warrants%20Valuation%20Method.pdf>.

²⁹ To match the Treasury's methodology, we compute the 10-year historical volatility using daily returns. We have 57 banks for which we are able to compute the 10-year historical volatility.

Table VII
TARP Warrant Subsidy

This table estimates the warrant subsidy provided by the Treasury to banks by computing the Volatility Subsidy, which is defined as the difference between a bank's 10-year historical volatility and the implied volatility of the Treasury's repurchase price for that bank. In the second column, we estimate the Volatility Subsidy using the implied volatility on the longest-dated call option covered on OptionMetric. *t*-Statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	Historical 10-Years	Implied Longest-Mat
Mean Volatility Subsidy	0.185*** (12.84)	0.103*** (6.34)
<i>N</i>	57	32

measure the effect of these volatility subsidies on warrant valuations using a binomial approximation of an American Call Option. These subsidies are presented on a bank-by-bank basis in Table VIII.³⁰

Overall, our results document three key findings: (i) banks received significant subsidies compared to private benchmarks, (ii) prepayment of preferred equity without any penalty contributed to the subsidies, and (iii) warrant repurchases favored the financial institutions. We provide a bank-level summary of these estimates in Table VIII for the sample of 20 banks for which we have data on traded preferred equity in private markets.

D. Payout Decisions

We now examine which financial stakeholders benefited from TARP subsidies. In particular, we consider how banks receiving larger subsidies distributed these gains through dividend payments to common equity holders and compensation to the CEOs following the repayment of TARP. Investigating changes in these payouts is particularly apt since TARP prohibited any increase in dividends per share and CEO compensation until the principal was repaid. This gave banks an incentive to renegotiate and quickly repay TARP investments. Understanding the relationship between payouts and TARP renegotiation is also important from the perspective of financial stability and risk-taking in banks. For example, Meiselman, Nagel, and Purnanandam (2018) document that banks with high profits and payouts in good times perform worse during a crisis. Since postrenegotiation payouts occurred in relatively good states of the world, increased payouts during this period have important implications for the banking sector's performance in bad times in the future.

To test how banks change their payouts around repayments, we construct a bank-year panel of CEO compensation and dividend payouts from 2005 to 2015. We measure the outcome variables, *CEO Payout* and

³⁰ We have shared a data set detailing the subsidies on a bank-by-bank basis for the full sample of banks on our websites as well as in the supplementary materials to this article.

Table VIII
Bank-Level Subsidy Estimates

This table provides a bank-level summary of the subsidy estimates for the sample of 20 banks with traded preferred equity in private markets. TARP Amt. is the dollar amount of the CPP investment in (\$) billions. IRR Preferred is the annualized return on the same banks' preferred equity over the same investment horizon as the Treasury's TARP investment. IRR Treasury is the annualized IRR of the cash flows that banks pay the Treasury. Subsidy (% Difference) is the difference between the two IRRs. Subsidy (\$Billions) is the NPV of the banks' cash flows paid to the Treasury discounted by the cumulative return on the same bank's preferred equity. Repayment Subsidy is the NPV of the remaining TARP cash flows at the time a bank repaid TARP divided by the outstanding principal amount. The repayment subsidy NPV is computed using the prevailing preferred equity yield as the discount rate. Warrant Subsidy is the difference between TARP repurchase price of a bank's warrants and the estimated dollar value from an American Call Option using a binomial approximation and the bank's unwinorsized 10-year historical volatility. We scale the dollar difference by the total TARP investment amount. Sold at Auction indicates that bank's warrant was sold to market participants rather than repurchased by the bank.

Bank Name	TARP Amt \$B	IRR Preferred	IRR Treasury	Subsidy %	Subsidy (\$Billions)	Repayment Subsidy	Warrant Subsidy
CITIGROUP INC.	25.0	38.08%	14.16%	23.92%	8.86	Converted	Sold (Auction)
JPMORGAN CHASE	25.0	35.77%	10.95%	24.82%	3.52	0.02%	Sold (Auction)
BANK OF AMERICA	25.0	24.10%	6.31%	17.79%	2.35	1.38%	Sold (Auction)
WELLS FARGO	25.0	12.51%	7.99%	4.52%	1.65	1.65%	Sold (Auction)
MORGAN STANLEY	10.0	83.68%	20.50%	63.18%	2.89	0.02%	3.71%
GOLDMAN SACHS	10.0	83.53%	23.16%	60.37%	2.44	0.02%	0.20%
PNC FINANCIAL	7.6	67.54%	8.91%	58.62%	3.14	2.81%	Sold (Auction)
U.S. BANCORP	6.6	32.23%	8.84%	23.39%	0.35	0.94%	0.14%
SUNTRUST BANKS	4.9	26.52%	5.38%	21.15%	1.72	1.72%	Sold (Auction)
CAPITAL ONE	3.6	83.63%	12.10%	71.54%	1.17	0.94%	Sold (Auction)
FIFTH THIRD BANCORP	3.4	62.38%	8.92%	53.46%	1.85	2.27%	5.04%
BB&T CORP.	3.1	70.34%	8.89%	61.45%	0.79	0.94%	0.92%
BNY MELLON	3.0	21.15%	12.46%	8.69%	0.10	0.02%	1.25%
KEYCORP	2.5	42.71%	6.27%	36.44%	1.20	1.72%	3.26%
STATE STREET	2.0	8.20%	10.00%	-1.80%	0.00	0.02%	0.86%
ZIONS BANCORPORATION	1.4	30.08%	5.27%	24.81%	0.76	0.00%	Sold (Auction)
HUNTINGTON BANK	1.4	51.23%	6.76%	44.47%	0.71	2.02%	3.98%
WEBSTER FINANCIAL	0.4	70.00%	7.75%	62.25%	0.22	2.03%	Sold (Auction)
CITY NATIONAL	0.4	109.78%	9.02%	100.76%	0.24	2.38%	1.77%
SVB FINANCIAL GROUP	0.2	29.83%	7.91%	21.92%	0.06	2.39%	0.03%

Dividend Payout using a bank's total CEO compensation and dollar amount of dividend payments scaled by the levels of these variables in the pre-crisis years of 2002–2007. Therefore, the dependent variables measure the level of payouts in a year relative to the average value in the precrisis period.

Banks repaid TARP in different years. We therefore use a dynamic event-study specification since using a static difference-in-differences specification when units have different treatment timing may result in severe bias (De Chaisemartin and d'Haultfoeuille (2020), Goodman-Bacon (2021), Sun and Abraham (2021), Borusyak, Jaravel, and Spiess (2024)).³¹ In the dynamic specification, we create indicator variables for every year relative to the year of TARP repayment by the bank. For instance, *Repayment Year* = + 1 equals one in the year after TARP repayment, zero otherwise. We then split the sample into banks with above and below median *Investment NPV* (i.e., TARP NPV on a per-dollar basis) to test whether banks with larger subsidies distributed higher payouts to their stakeholders.

Table IX presents the results. Columns (1) and (2) contain the results for CEO compensation. CEO payouts increased considerably for the high-subsidy group for a period of three years after the repayment. In contrast, banks receiving low subsidies experience no increase in CEO compensation. Columns (5) and (6) similarly show that the high subsidy banks significantly increased their dividend payouts to shareholders in the years following TARP repayment.

Our earlier results show that banks that were successful in repurchasing their warrants from the Treasury received significant subsidies. We next test whether CEOs were able to capture some rents from these renegotiations. Columns (3) and (4) of Table IX present the results for CEO payouts across banks that were able to repurchase their warrants versus banks whose warrants were sold in the market. CEO compensation for banks that repurchased their warrants from the Treasury consistently increased by a significant amount, ranging from 51% to 93% of precrisis levels, in the years following the repayment. In contrast, for banks whose warrants were sold in the market, CEO compensation did not increase. The evidence supports the view that bank managers captured some of the rents provided by warrant repurchase subsidies.

E. Additional Analyses and Robustness Tests

We next conduct a series of robustness tests and additional analyses to rule out some alternative explanations and to provide additional economic insights into our results.

³¹ We estimate the dynamic model using Ordinary Least Squares (OLS) in the paper, but in the Internet Appendix, we show that the results are robust to using the Sun and Abraham (2021) estimator, which accounts for the presence of heterogeneous treatment effects.

Table IX
Financial Stakeholder Payouts

This table examines changes in financial stakeholder payouts around TARP repayments sorted by the TARP subsidy a bank received. The panel consists of bank-year observations from 2005 to 2015 for all banks covered by Bank CRSP-Compustat. Samples are split into Low Subsidy and High Subsidy based on whether a bank's NPV/Investment is above or below the sample median. In columns (1) to (4), the outcome variable is *CEO Payout*, which is defined as annual CEO compensation scaled by the average level of CEO compensation in the precrisis years of 2002 to 2007. In columns (5) and (6), the outcome variable is *Dividend Payout*, which is defined as annual dividends scaled by the average level of dividends payments in the precrisis years 2002 to 2007. Outcome variables are winsorized at the 2% level. *Repayment Year = 0* is an indicator variable equal to one in the year that a bank repaid the TARP principal in full. *Repayment Year = +1* equals one in the year after a bank repaid TARP, and so on. Standard errors are clustered at the bank level. *t*-Statistics are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	CEO Payout				Dividend Payout	
	Low Subs. (1)	High Subs. (2)	Sold (3)	Repurchased (4)	Low Subs. (5)	High Subs. (6)
Repayment Year = -3	0.109 (0.56)	-0.047 (-0.66)	0.300* (1.82)	-0.223* (-2.01)	-0.094 (-0.81)	0.014 (0.15)
Repayment Year = -2	-0.020 (-0.17)	-0.044 (-0.56)	0.135 (1.10)	-0.076 (-0.80)	-0.168* (-1.83)	0.236* (1.75)
Repayment Year = 0	0.023 (0.11)	0.595*** (3.30)	-0.002 (-0.01)	0.515*** (2.78)	-0.268 (-1.08)	0.264** (2.04)
Repayment Year = +1	-0.129 (-0.48)	0.672*** (3.13)	-0.188 (-0.76)	0.604*** (3.11)	-0.386 (-1.14)	0.633*** (3.00)
Repayment Year = +2	-0.179 (-0.51)	0.956** (2.59)	-0.528 (-1.53)	0.774** (2.69)	-0.197 (-0.47)	1.039*** (3.48)
Repayment Year = +3	-0.379 (-0.83)	0.947* (1.98)	-0.736 (-1.48)	0.756** (2.03)	-0.433 (-0.82)	1.194*** (3.00)
Repayment Year ≥ 4	-0.251 (-0.39)	1.335* (1.89)	-0.624 (-0.79)	0.936* (1.79)	-0.538 (-0.79)	2.193*** (3.00)
Observations	421	341	220	397	1,062	1,027
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.585	0.599	0.606	0.617	0.508	0.480
Within R ²	0.007	0.060	0.033	0.046	0.006	0.035

E.1. Benchmark Returns and Policy Uncertainty

As discussed in Section II, market participants faced two types of risk during the early days of the crisis: uncertainty about the policy intervention itself and conditional on the resolution of this uncertainty the risk of investing in financial institutions with high default risk. Our subsidy measure may therefore capture both of these effects. While we cannot directly decompose the subsidy into these two parts, we conduct a test that measures the private-market return after uncertainty with respect to the policy actions and, in turn, aggregate stability of the financial sector had diminished. In this test, we compare TARP's return with the return earned by a market investor in the preferred

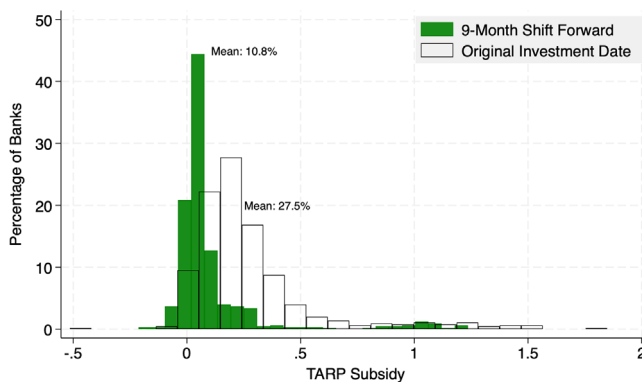


Figure 5. TARP benchmark adjusted for policy uncertainty. This figure plots a histogram distribution of the TARP subsidy (i.e., the difference between the annualized rate of return from investing in a bank's preferred equity and the TARP return) using an alternative benchmark that shifts the investment horizon of the benchmark by nine months from the date of the original investment. (Color figure can be viewed at wileyonlinelibrary.com)

equity of the banks six months, nine months, or one year after the TARP investment date.

Figure 5 plots the distribution of subsidy using the nine-month-shifted version of the benchmark return. On average, taxpayers earned a 10.8% lower return compared to the benchmark return shifted by a period of nine months. Table IA.VI of the [Internet Appendix](#) presents results of this analysis for all three shifted returns, that is, benchmark returns shifted by 6, 9, and 12 months. Even for the most conservative one-year benchmark, banks received a 6% subsidy relative to the private market. This analysis indicates that TARP investments earned a lower return than a private-market investor who invested in these banks at a time when concerns about the uncertainty surrounding the policy interventions and aggregate stability of the banking sector had been resolved for all practical purposes.

E.2. Liquidity of Benchmark Returns

Financial markets did not function smoothly soon after the failure of Lehman Brothers. Therefore, one may be concerned that our benchmark return does not represent an investible strategy for private-market investors. To address this issue more directly, we compute the average bid-ask spread of preferred equity prices during the crisis period. We plot the evolution of half of the bid-ask spread on these securities in Figure IA.5 of the [Internet Appendix](#), as they represent the trading costs an investor would incur in purchasing the security (Stoll (2000), Bao, O'Hara and Zhou (2018)). As expected, the half-spread widened from a level of 25 bps during the precrisis period to a peak of 170 bps at the time of Lehman's failure. At the time of CPP investments, however, the spread came down to a level of 40 bps. Even if we take the peak full spread of

3.4% as the representative cost of transactions for the benchmark securities, it is significantly lower than our subsidy estimate of 30%. Note that our benchmarking exercise based on the Warren Buffett deal in Goldman Sachs does not suffer from this critique since the benchmark returns were also from a privately placed investment. Finally, our benchmarks based on shifted returns, that is, benchmark returns shifted by 6 to 12 months, further ameliorates concerns about lack of liquidity affecting our results because by that time, the markets were functioning reasonably smoothly.

We next collect evidence from existing literature on the economic magnitude of returns that investors earn due to such market breakdown. Pasquariello (2014) constructs a measure of market dislocation based on the violation of standard arbitrage conditions in equity, foreign currency, and money markets during the 2008 crisis. He shows that stocks more sensitive to the dislocation, that is, stocks that are more likely to be affected by the market breakdown, earn a higher expected return. In terms of economic magnitude, the difference in annualized abnormal return is about 5.3% per annum between stocks that fall in the bottom versus top decile of this beta. Our estimated subsidy of 30% per annum is too large to be explained by even the differences in expected return across extreme portfolios documented by these studies.

E.3. Realized versus Expected Returns

Another potential concern with our empirical analysis can come from the difference in realized and expected returns on our benchmark securities. Specifically, our returns are based on one possible realization of the states of the world. If our replication exercise were perfect, there would be no concern about the subsidy calculation. However, the presence of warrants makes the replication exercise imperfect and hence raises concern: could the TARP's realized return on warrants be much higher under a different realization, biasing our subsidy calculation upward? Three key features of our analysis make this concern a relatively minor one. First, our realized returns come from a relatively good state of the world, and hence, the warrant payoffs are already large. Second, the fact that warrant terms were renegotiated ex post suggests that their payoff is capped even in more attractive states of the world. Finally, as shown by Veronesi and Zingales (2010), warrants comprised a relatively modest part of the investment based on ex-ante valuations.

E.4. Sensitivity Analysis

We perform a sensitivity analysis to see how TARP subsidies might change if upfront contract terms were set differently. We generate a series of counterfactual cash flows assuming that either the preferred equity coupon was larger or the Treasury received additional warrant shares. We then recompute NPV by discounting these cash flows holding constant the capital market benchmarks. Internet Appendix Table IA.VIII contains the results. For the NPV to become zero, that is, for the Treasury to break even on TARP, we find that the warrant

shares must be 80% to 100% of the principal amount holding all other terms of the contract fixed. Alternatively, the coupon rate on preferred equity must be as high as 40% holding all other terms fixed. As shown earlier, Warren Buffett's deal had a much higher coupon rate and warrant amounts attached to the investment, similar to the results of our sensitivity analysis.

V. Conclusions

Government bailouts happen in bad times, but the repayments occur in good times. At the time of investments, recipients have higher risk and risk-aversion in the economy is high as well. Therefore, from a purely financial perspective, firms raising capital at such a time should pay sufficiently high returns to compensate their investors for bearing this risk. Using returns on several benchmarks of comparable or lower risk than the preferred equity investments made under TARP, we show that recipients paid a much lower rate of return to the U.S. Treasury compared to the benchmarks. Contrary to the popular claims that TARP investments were "profitable" for taxpayers on a purely financial basis, our analysis shows that they were heavily subsidized on a risk-adjusted basis. In addition, ex-post renegotiation of TARP terms further contributed to the subsidy enjoyed by the banks. While a welfare calculation is beyond the scope of our paper, the measurement of TARP subsidies is an essential first step in that direction.

Our findings have important implications for the design of future bailouts, in terms both of ex-ante terms of the contract and ex-post renegotiation. In TARP's case, the renegotiation of the contract benefited bank shareholders at the expense of taxpayers. Since these renegotiations happened in relatively better states of the world, it is not clear that relaxing the contract terms in favor of the recipients has any value for financial stability. Instead, our findings suggest that the relaxation of contract terms helped banks pay high dividends, which weakens their capital position further. These inputs should be helpful for the design and implementation of future bailouts.

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Appendix

A. Example Calculations

JP Morgan received a TARP preferred equity investment of \$25 billion on October 28, 2008. As per the TARP CPP term sheet,³² the original (earliest) repayment date was October 28, 2011. JP Morgan repaid the TARP investment in full, however, on June 17, 2009. In addition to returning the full investment amount of \$25 billion, JP Morgan paid 5% per annum cumulative

³² Available at <http://www.treasury.gov/press-center/press-releases/Documents/document5hp1207.pdf>

quarterly dividends (as per the term sheet), resulting in \$795 million in dividend payments to the Treasury. The preferred equity investment also granted the Treasury 88,401,697 warrant shares (15% of the preferred equity investment) to purchase new issues of JP Morgan stock on October 28, 2018 at an exercise price of \$42.24 (the 20-day moving average of the stock price around the initial investment amount). On December 10 2009, the Treasury sold these warrants to market participants at auction for a total of \$950 million or \$10.75 a share. The annualized rate of return from the TARP investment in JP Morgan is 10.95%.

To calculate a benchmark return for JP Morgan, we compute the annualized return of investing in a market index from October 28, 2008 to December 10 2009, the investment horizon of the TARP preferred equity investment. Investing in the S&P US Preferred Stock Index over this horizon would have earned an annualized return of 48.96%. The benchmark index subsidy is simply the difference from the TARP return, 38.0%. Investing in JP Morgan preferred equity over the same time horizon would have earned an annualized return of 35.77%. The same bank subsidy is therefore 24.82%.

A.1. TARP Loss Example

It is possible that TARP makes a negative return on its investment. This happens if a bank fails to repay the preferred equity principal and/or TARP sells the preferred equity at a loss to market participants. An example is Northern States Financial Corporation, which received a \$17.2 million investment from TARP in 2009. After five years, TARP sold the preferred equity at a loss for \$6 million. Including dividend payments, the total cash back from this investment was only \$6.42 million. The warrants became worthless and generated no proceeds.

In some cases, TARP effectively earns an annualized IRR of -100% . One example of this is Midwest Banc Holdings, which closed and went into receivership of the Federal Deposit Insurance Corporation (FDIC). The IRR was effectively -100% .

A.2. Duration Risk

A potential concern with the analysis so far is that traded preferred equity securities may not be directly comparable to TARP investment due to differences in the duration of these investments. The weighted-average life of the TARP investments in our same bank preferred equity sample is 1.18 years. The preferred equity used as the benchmark, however, has an average life of 43.35 years.³³ Since these two investments have a difference in duration of 42.17 years, the estimated subsidy might reflect differences in returns due to duration risk. We check the extent to which duration risk affects the subsidy

³³ For perpetual preferred equity, we assume a life of 50 years. Since many of these preferred equity securities are callable, the effective maturity may be shorter.

estimate by calculating the return to a 30-year Treasury bond over the same 1.18-year period as our same bank preferred equity investments.³⁴ We find an annualized return of -3.39% on 30-year Treasury bonds over this period. This result indicates that the previously reported subsidy measure would, in fact, be larger once accounting for the differences in duration risk between the TARP investment and traded preferred equity during this sample period.

B. Construction of the Replicating Portfolio Benchmark

Our replicating portfolio invests 90% in the bank's plain preferred equity and 10% in the Warrants. Preferred equity returns are computed as discussed in the paper. To estimate the return on the warrant's part of the replicating portfolio, we use a valuation model that closely resembles the Treasury's valuation methodology to estimate the fair market value of the warrants each day from the date of TARP investment to the day of warrant disposal. Based on these fair values, we compute daily returns to the warrants. The benchmark return is simply the weighted average return of the portfolio.

Warrants are valued as an American Call Option on a dividend paying stock. Following the Treasury's model, we use a binomial approximation to the Black-Scholes model for this valuation. The estimation of the model requires a number of assumptions on the inputs as well as valuation technique used to get these estimates. We closely follow the methodology of the Treasury in making these assumptions and modeling choices. Below we list our assumptions, and the underlying rationale, for each input of the model.³⁵

- **Stock Price (S):** We obtain the daily stock price from the CRSP data set. We adjust the stock price as well as the variables below to account for any corporate actions/stock splits.
- **Strike Price (K):** At the time of the investment, the strike price for warrants was set at a 20-day moving average of the company's common stock's price measured two weeks before the investments. The strike price was adjusted downward following any dividend payment amount in excess of the dividend per share at the time the strike price was set. We follow the same methodology to compute the strike price for each bank's warrant.
- **Risk-free rate (r):** The Treasury uses the yield on a risk-free bond that matches the remaining maturity of the warrant. We use the risk-free rate on 10-year Treasury as our input for r , since most of these warrants had a remaining maturity of 7 to 10 years during our computation period.
- **Dividend Payout (y):** We consider the annual dividend yield computed using the bank's most recent quarterly dividend payment and the stock price on a given day.

³⁴ A 30-year Treasury bond is the closest available market benchmark to capture the 42.17-year difference in duration of the two investments.

³⁵ Available at <https://www.treasury.gov/initiatives/financial-stability/TARP-Programs/bank-investment-programs/cap/Pages/default.aspx>.

- Volatility (σ): The Treasury uses both the historical volatility and implied volatility to construct a 10-year forward volatility curve, and takes the average of these volatilities as the input in the valuation model. The initial part of the forward volatility curve comes from the implied volatility of the traded option; the latter part comes from 10-year average historical volatility. In our valuation model, we use the implied volatility for the longest dated traded option available in the OptionMetric database. We compute historical volatility using common equity returns from CRSP. Our base model uses the average of the two volatilities as the input for σ .
- Dilution effects for warrants: Warrants are different from standard call options in the sense that if the warrants are exercised, the company has to raise new shares. The Treasury makes no adjustments to the Black-Scholes valuation model for the resulting dilution. We follow the same approach. The rationale behind this nonadjustment is that the stock market rationally anticipates these effects and hence the current stock price of the company, a key input in the valuation model, already accounts for this effect.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Appendix S1: Internet Appendix.
Replication Code.

